Natalie Ko | ko12nat@gmail.com | 909.753.4247 | linkedin.com/in/ko12nat/

EDUCATION

Harvey Mudd College, Claremont, California, USA — Expected graduation: May 2028
B.S., Engineering | Major GPA 4.0/4.0 | Cumulative GPA 3.918/4.0 | Dean's List (all graded semesters)

Relevant coursework: Microprocessor Systems: Design & Application; Continuum Mechanics; Electrical & Magnetic Circuits/Devices; Digital Electronics & Computer Engineering; Advanced Systems Engineering; Applied Math for Engineering; Experimental Engineering; Intro Engr Design/Manufacturing

AWARDS AND ACHIEVEMENTS

Tau Beta Pi Member (United States Engineering Honours Society, Top 12.5% of Junior Class) — 2026 - Present
Harvey Mudd Merit Scholarship (Top 15% of Class) — 2024 - Present

WORK EXPERIENCE

R&D Manufacturing Contractor, MiniMed (Harvey Mudd College Clinic Program) — Aug 2026 - Present

- Optimizing Continuous Glucose Monitor sensor manufacturing process through reducing cycle time
- Prototyped & tested a sensor clamping fixture, prevented sensor damage & improved operator ergonomics
- Performed materials analysis & testing, met ISO 7 classifications & withstood corrosive baths during 24/5 use

Engineering Intern, Cathay Pacific, Hong Kong — Jul 2026 - Aug 2026

- Developed a cross-company maintenance Power App; cut daily task time from 1hr+ to 24min (100% adoption)
- Built a Power BI automated predictive model, optimized servicing for 13 cabin layouts under operational limits
- Co-developed an Ultralytics YOLO computer vision model to detect cabin defects & automate inspections

Student Researcher, Flow Imaging Lab at Mudd, Dept. of Engineering, Harvey Mudd College — Jan 2026 - Present

- Designed & manufactured a mechanical 3-axis calibrator for a live class-4 laser for Particle Image Velocimetry
- Developed a PI-controlled degree of freedom (fin) with op-amp circuits for high-res potentiometer feedback
- Processed 6-axis load cell data using MATLAB to quantify thrust generated by a biomimetic fish fin

Lab Teaching Assistant / Lab Improver, Intro Engr Design/Manufacturing, Harvey Mudd College — Aug 2025 - Present

- Taught 20+ students SolidWorks, CAD, motion studies, finite element analysis, prototyping, material selection
- Reviewed technical drawings for manufacturing; final products used in research labs, courses, school services
- Authored 5 SolidWorks tutorials teaching feature logic and software troubleshooting, used by 60+ students
- Pioneered new student role, eliminated continuity errors between teachings & deliverable expectations

Repair Steward, Makerspace, Harvey Mudd College — Jan 2025 - Present

- Maintained, repaired, & troubleshot fabrication equipment failures; minimized downtime for 2500+ users
- Coordinated with 5 undergraduates to write/update user, safety, & repair manuals & SOPs for all equipment
- Optimized services; provided remote technical support, analyzed maintenance logs, reported inconsistencies

Student Researcher, HMC Biophotonics Lab, Dept. of Engineering, Harvey Mudd College — Jan 2025 - Dec 2025

- Co-developed an open-source in vivo imaging system, reduced production cost to <1% of commercial price
- Improved speckle contrast & bioluminescence imaging capabilities; device used by Rathbun Lab @ CMC
- Devised experimental optics setups; troubleshot modules with oscilloscopes; processed images with Python

PROJECT EXPERIENCE

Keypad Scanner, Microprocessor Systems: Design & Application, Harvey Mudd College — Aug 2026 - Sept 2026

- Implemented an FPGA design to read & display last 2 4x4 matrix keypad inputs on a dual 7-segment display
- Utilized communicating finite state machines to account for switch bounce, time-multiplexed display etc.

Team Leader, Ocean Debris-detecting Robot, Experimental Engineering, Harvey Mudd College — Jan 2026 - May 2026

- Led 3 undergraduates to design & deploy an autonomous underwater vehicle for seafloor debris mapping
- Integrated sensors (magnetometer, thermistor) with Teensy 4.0 to collect data via ADCs, SPI & I2C protocols
- Programmed P-control loops in C++ for surface & depth navigation from pressure sensor & GPS readings
- Designed custom op-amp scaling & filtering circuits to maximize sensor precision across input voltage range

Team Leader, Abyss Explorer, Intro Engr Design/Manufacturing, Harvey Mudd College — Feb 2025 - May 2025

- Led 4 undergraduates to fabricate remote systems for environmental data collection in a radioactive mine
- Created SolidWorks parts, assemblies, technical drawings, & BOM for manufacturing & client reference
- Managed project deadlines & a $500 budget; client successfully deployed the final product in fall 2025

PUBLICATIONS AND PRESENTATIONS

Poster/Presenter: "Integrating Laser Speckle Contrast Imaging with OpenIVIS," **Ko, N.**, Tran, M., Brake, J. Southern California Conference for Undergraduate Research, Nov 2025